

Announcements

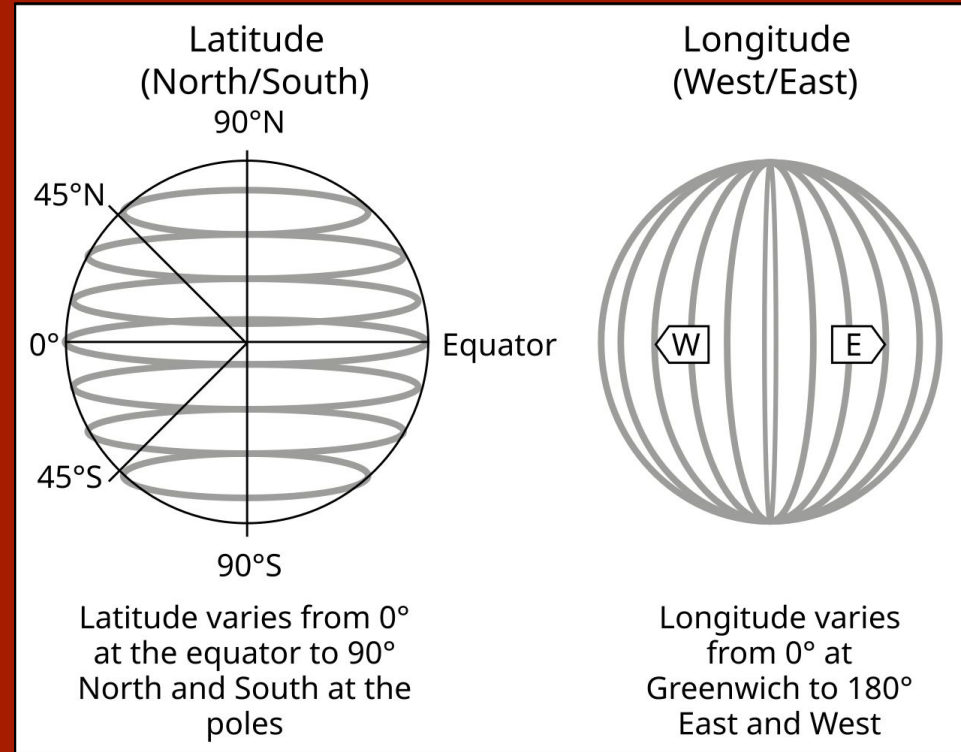
- Please double check that you are in the correct class - there have been several students marked absent because they went to the wrong one
 - We have 2 labs meeting each day, in different class rooms, with different TAs
 - Check the class number on Canvas or myasu - it should be a 5-digit number starting with 70, you can ask the TA if you are not sure
 - You can also check which TA you are getting Canvas announcements from (barring the occasional one that gets sent to all sections) - make sure that is the same one teaching the lab you go to
- Next week will also be a night lab, and it will be our first observing lab
 - (We will actually be looking at stars)

Coordinate Systems

How to figure out where things are

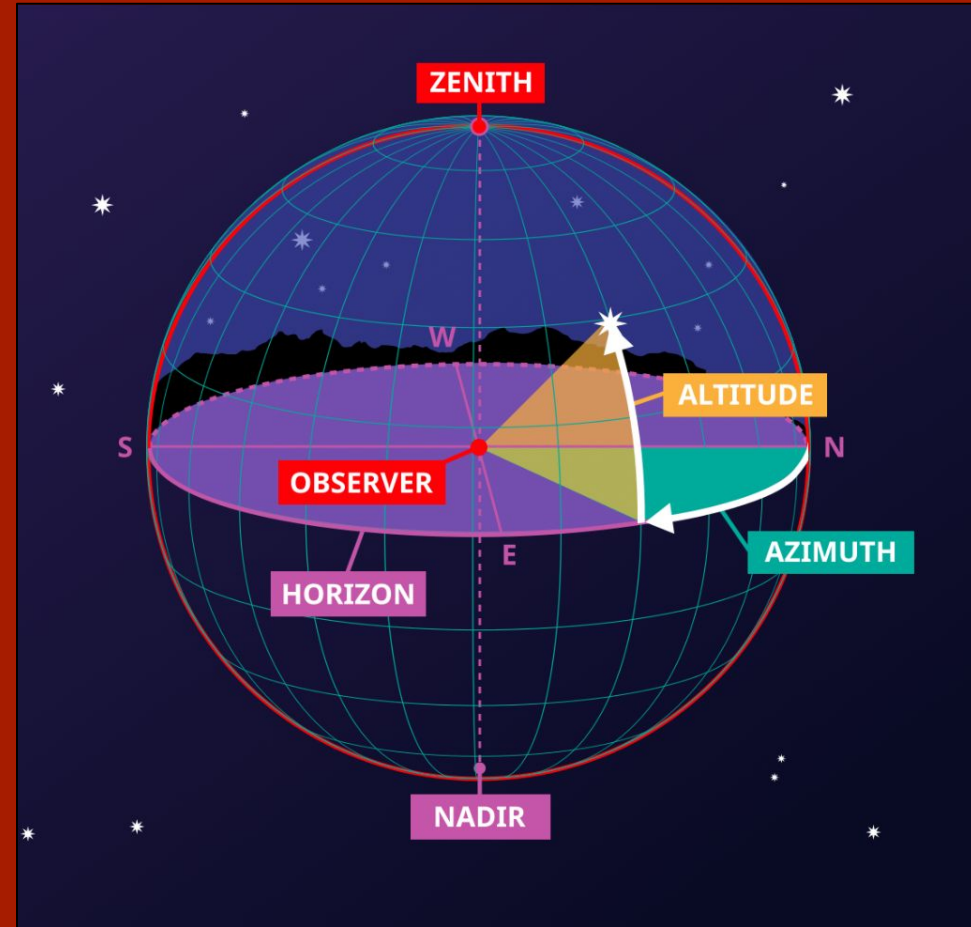
Geographic System

- Locations on the Earth's Surface
- Latitude is horizontal lines parallel to the equator
 - Two ways to help you remember:
“flatitude” because they are horizontal
 - Or “ladderude”, because they are like the rungs of a ladder
- Longitude is vertical lines, perpendicular to the equator
- Measured in degrees ($^{\circ}$), arcminutes ($'$), and arcseconds ($''$)



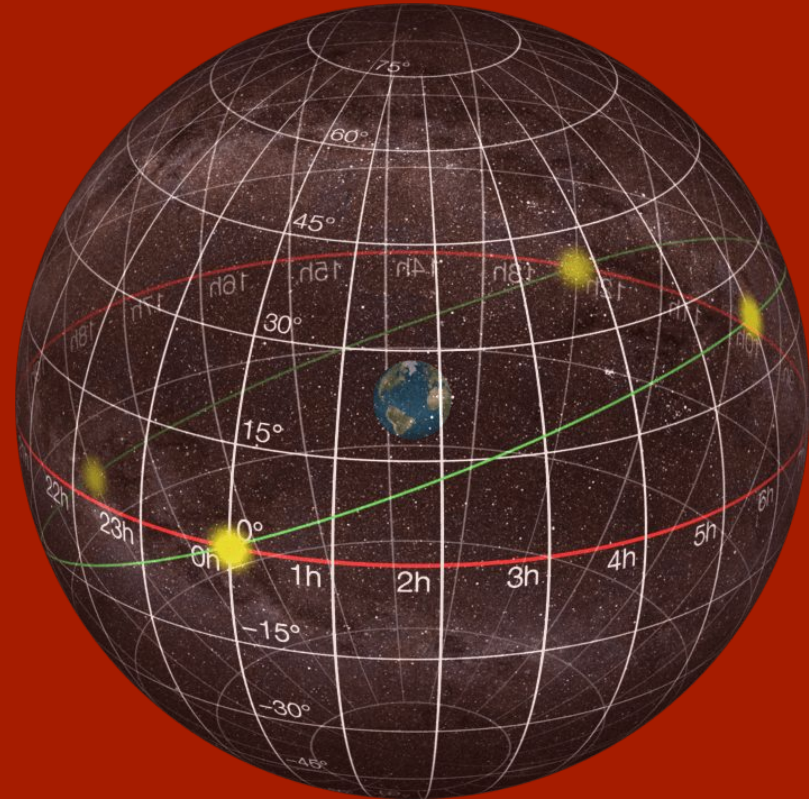
Horizon System (Alt-Az)

- One of the systems for locations in the sky
- “Local” to the observer
 - Meaning it is different depending on where you are on the Earth
- Changes over time
 - Due to the rotation of the Earth
 - Things appear to rotate around Polaris (the north star)
- Measured in degrees ($^{\circ}$), arcminutes ($'$), and arcseconds ($''$)



Equatorial System (Right Ascension (RA) and Declination (Dec))

- Also used for the sky
- Based on the Celestial Equator (Earth's equator projected into space) and Earth's north and south poles
- Also important is the Ecliptic - the plane of Earth's orbit around the Sun
- Measured in hours, minutes, seconds (RA) and degrees (Dec)
- Positions of objects are fixed* as opposed to the Horizon System's movement
 - *Technically, everything is moving. The further they are from Earth, the longer it takes for the change in RA/Dec to be significant - Planets take ~1 day, stars take millenia



Today's Lab (Part 1)

- Answer question 1-9 based on the different coordinate systems
- Then questions 10-17 to practice using the coordinate systems on the star Kochab and the unknown object from the DSL VR Experience
- Then questions 18-21 to determine how the unknown object moves
- Question 22-27 to determine which coordinate system is best for the different parts of this project
- Plot the objects movement on a star chart - and use that to answer questions 29-33

Today's Lab (Part 2)

- Answer questions 34-38 to identify which observatories may have unknowingly imaged the object before its discovery
- Open Stellarium Web (<https://stellarium-web.org/>)
- Look around and get familiar with the controls
- Input the location (bottom left) and date/time (bottom right) of each observatory's observations
- Check that the object would have been high enough above the horizon, and answer questions 39-40



Image Credits (In Order of Appearance)

- Wikipedia (Geographic Coordinate System)
 - https://en.wikipedia.org/wiki/Geographic_coordinate_system
- IAU OAE
 - <https://astro4edu.org/resources/diagram/4g46V657yk21/>
 - Cropped by me
- Wikipedia (Equatorial Coordinate System)
 - https://en.wikipedia.org/wiki/Equatorial_coordinate_system
- Screenshot of <https://stellarium-web.org/>, taken by me
 - Location: 33.43220 -111.93430 and date/time: 9/9/26 at 19:51:59 local time